Chapter-3

Conservation of Linear Momentum and Collision

Textbook: Fundamentals of Physics by Halliday, Resnick & Walker [extended 10th edition]

Topics to be Covered:

9-4: Linear Momentum. 9-5: The Linear Momentum of a System of Particles. 9-7: Conservation of Linear Momentum. 9-8: Momentum and Kinetic Energy in Collision. 9-9: Inelastic Collision in One Dimension. 9-10: Elastic Collision in One Dimension.

Theory

1. Define (a) linear momentum for a system of particles (b) elastic and inelastic collision.
2. Briefly explain center of mass.
3. State and explain the Law of Conservation of Linear Momentum.
4. Derive the expression for the velocity of two colliding objects that are stick together after a completely inelastic collision for stationary target.
5. Derive the expressions for the velocities of two colliding objects after an elastic collision in one dimension for stationary target.
6. Derive the expressions for the velocities of two colliding objects after an elastic collision in one dimension for moving target.

Linear Momentum (Chapter-9)

1. [18] A 0.70 kg ball moving horizontally at 5.0 m/s strikes a vertical wall and rebounds with speed 2.0 m/s. What is the magnitude of the change in its linear momentum?

2. [19] A 2100 kg truck traveling north at 41 km/h turns east and accelerates to 51 km/h. (a) what is the change in the truck's kinetic energy? What are the (b) magnitude and (c) direction of the change in its momentum?

3. [91] A 75 kg man rides on a 39 kg cart moving at a velocity of 2.3 m/s. He jumps off with zero horizontal velocity relative to the ground. What is the resulting change in the cart's velocity, including sign?

4. [39] A 91 kg man lying on a surface of negligible friction shoves a 68 g stone away from himself, giving it a speed of 4.0 m/s. What speed does the man acquire as a result?

5. [42] An object, with mass m and speed v relative to an observer, explodes into two pieces, one three times as massive as the other; the explosion takes place in deep space. The less massive piece stops relative to the observer. How much kinetic energy is added to the system during the explosion, as measured in the observer's reference frame?

Inelastic Collision in One Dimension (Chapter-9)

6. [50] A 5.20 g bullet moving at 672 m/s strikes a 700-g wooden block at rest on a frictionless surface. The bullet emerges, traveling in the same direction with its speed reduced to 428 m/s.(a) What is the resulting speed of the block? (b) What is the speed of the bullet–block center of mass?

7. [55] A 5.0 kg block with a speed of 3.0 m/s collides with a 10 kg block that has a speed of 2.0 m/s in the same direction. After the collision, the 10 kg block travels in the original direction with a speed of 2.5 m/s. (a) What is the velocity of the 5.0 kg block immediately after the collision? (b) By how much does the total kinetic energy of the system of two blocks change because of the collision?

8. [54] A completely inelastic collision occurs between two balls of wet putty that move directly toward each other along a vertical axis. Just before the collision, one ball, of mass 3.0 kg, is moving upward at 20 m/s and the other ball, of mass 2.0 kg, is moving downward at 12 m/s. How high do the combined two balls of putty rise above the collision point? (Neglect air drag.)

Elastic Collision in One Dimension (Chapter-9)

9. [61] A cart with mass 340 g moving on a frictionless linear air track at an initial speed of 1.2 m/s undergoes an elastic collision with an initially stationary cart of unknown mass. After the collision, the first cart continues in its original direction at 0.66 m/s. (a) what is the mass of the second cart? (b)What is its speed after impact?

10. [62] Two titanium spheres approach each other head-on with the same speed and collide elastically. After the collision, one of the spheres, whose mass is 300 g, remains at rest. (a) What is the mass of the other sphere? (b) What is the speed of the two-sphere center of mass if the initial speed of each sphere is 2.00 m/s?

11. [63] Block 1 of mass m_1 slides along a frictionless floor and into a one-dimensional elastic collision with stationary block 2 of mass $m_2=3m_1$. Prior to the collision, the center of mass of the two-block system had a speed of 3.00 m/s. Afterward, what are the speeds of (a) the center of mass and (b) block 2?

12. [65] A body of mass 2.0 kg makes an elastic collision with another body at rest and continues to move in the original direction but with one-fourth of its original speed. (a) What is the mass of the other body? (b) What is the speed of the two-body center of mass if the initial speed of the 2.0 kg body was 4.0 m/s?